Learn Embedded Systems

This document is divided into two sections to provide comprehensive coverage of embedded systems and hardware engineering. The first section serves as an overview, focusing on essential concepts related to embedded systems, including Arduino and similar microcontrollers (MCUs). This section aims to provide a foundational understanding of embedded systems, their functionalities, and their applications.

The second section delves into more advanced topics, covering a wide range of aspects within embedded systems and hardware engineering. This section offers in-depth exploration of subjects such as microcontroller architectures, real-time operating systems, embedded software development, hardware design principles, circuitry, and electronic components. By delving into these areas, the document aims to provide a comprehensive understanding of embedded systems and equip readers with the knowledge necessary to engage in hardware engineering endeavors.

Overall, this document's structure ensures that both newcomers and experienced individuals in the field of embedded systems can benefit from its content. The initial section lays the groundwork, while the subsequent section delves deeper into the intricacies of embedded systems and hardware engineering, catering to a wider range of readers with varying levels of expertise.

==========

Follow US

www.linktr.ee/electro.sc

==========

Please always refer to this LINK in order to get the latest update to this doc!
Check out my resources HERE!
If you want to be notified with updates, please leave your email in the comment section.

Section 1:

Outline:

1) Essential Programming Skills: 2
2) Arduino and Electronics Basics: 2
3) Beyond Arduino: 3
4) Other Resources: 4

1) Essential Programming Skills:

To expand beyond the limitations of the basic Arduino framework, a solid foundation in programming, particularly in C/C++, is indispensable. Proficiency in fundamental concepts such as data types, functions, arrays, strings, and structs is crucial. A comprehensive grasp of pointers is necessary to efficiently manage memory-related operations. Familiarity with the fundamentals of the C preprocessor is also essential. Additionally, a strong understanding of Object-Oriented Programming (OOP) is vital for comprehending the design principles of C++ libraries and effectively utilizing them. The following resources are recommended for acquiring the knowledge and skills outlined below:

- C++ Playlist
- Pointers in C++ Playlist (Mandatory)
- C Playlist
- Bitwise operators in C
- Standard integer library
- C preprocessor

2) Arduino and Electronics Basics:

To transcend the capabilities of Arduino, it is essential to possess a proficient command over the Arduino framework. In this regard, I highly recommend exploring the comprehensive resources available. Firstly, THIS PLAYLIST provides an invaluable collection of tutorials and demonstrations to enhance your understanding and proficiency with Arduino.
Additionally, this book C Programming for Arduino offers a comprehensive guide to mastering C programming specifically tailored for Arduino applications.

To consolidate your knowledge, it is highly encouraged to engage in numerous hands-on projects. Through practical implementation, you will gradually advance your skills and reach a stage where you crave additional functionality from your microcontroller. Emphasizing project-based learning will allow you to not only reinforce your existing abilities but also identify the specific areas where you require further development.

3) Beyond Arduino:

- FSMs and simulating Multitasking (no spaghetti code)
- Levelup Arduino code: Registers
- Levelup Arduino code: Timers/Counters
- Levelup Arduino code: External Interrupts
- Bootloaders and programmers
- Improve your Arduino skills playlist
- Arduino BareMetal Playlist
- Going Beyond Arduino playlist
- Read the datasheet !!!
- Preprocessor and Volatile Keyword
- RTOS (real-time operating systems) basics

To effectively comprehend the contents of the ATMEGA328p or any other microcontroller's reference manual, it is crucial to establish a strong foundation in **Digital Systems**. Additionally, acquiring knowledge about the **basics of Computer Architecture** is highly recommended.

To gain expertise in these subjects, I enthusiastically suggest exploring the informative playlist available on the YouTube channel of Ben Eater. THIS PLAYLIST offers comprehensive tutorials and explanations that cover essential concepts in Digital Systems and Computer Architecture.
By engaging with this resource, you will enhance your understanding of the underlying principles and mechanisms that govern microcontrollers and their functioning. This will enable you to navigate through reference manuals with confidence and effectively comprehend the technical details they present.

- Begin by studying a chapter on a peripheral in the ATmega328p reference manual, configuring and testing it multiple times for proficiency.
- Transition to other microcontrollers like STM32 or ESP, following a similar process to broaden your knowledge and skills in embedded systems engineering.

4) Other Resources:

If you have developed a keen interest in learning and working with embedded systems, I'm delighted to provide you with additional resources to further nourish your curiosity. The following sources offer valuable insights and opportunities for continued exploration:

- Modern Embedded systems programming course (best embedded playlist ever, it is equivalent to 2 or 3 university courses!)
- Build systems like Make or better: CMake
- Sober YT channel (c programming, Operating systems and embedded)
- Embedded Systems - Shape the World - MOOC (Online Course offered by University of Texas at Austin)
- Embedded Systems Conference (ESC) (Embedded Systems conferences and workshops)
- Embedded Artistry blog (every other resource you need is here)
- Embedded.fm podcast (Podcasts with expert in the field)
- Battery University (Everything you need to know about batteries)

This section is made with the help of my friend **"Zakaria M."**, thanks for contributing!

Section 2:

Introduction:

This document serves as a curated collection of valuable resources for individuals interested in learning about embedded systems engineering. The resources included cover various aspects of embedded systems, providing a comprehensive learning experience for beginners and enthusiasts seeking to expand their knowledge and skills in this field.

The collection comprises a diverse range of resources, such as online courses, books, tutorials, conferences, and online communities. These resources offer a combination of theoretical knowledge, practical examples, hands-on projects, and opportunities for networking and collaboration. By exploring these resources, readers can gain a solid understanding of embedded systems engineering principles, develop proficiency in programming microcontrollers, and enhance their hardware design and development skills.

Whether you are starting from scratch or seeking to deepen your understanding of embedded systems, this document provides a valuable compilation of resources that cater to different learning preferences and levels of expertise. It serves as a guide for self-paced learning, enabling individuals to embark on a structured journey of acquiring the necessary knowledge and practical skills required for embedded systems engineering.

Let's begin our Embedded Systems journey with these resources:

[ ] Article: "Introduction", Barr Group
[ ] Video: "Embedded Systems Overview", YouTube

These resources will provide you with a solid foundation and a comprehensive overview of embedded systems. The article offers valuable insights and information, while the video provides visual demonstrations and explanations to enhance your understanding.

By reading the recommended article and watching the provided video, you will gain a clear idea and overview of embedded systems, equipping you with the necessary knowledge to begin your exploration in this field.

Now, let's jump to our resources and never skip a step!

Outline:

1) Development Environment: 9
- a) Cross Compilations: 9
- b) Development Tools: 10
  - i) IDE: 10
  - ii) Oscilloscopes and Logic Analyzers: 10

2) Embedded Hardware: 11
- a) Computer Architecture: 11
  - i) RISC: 11
  - ii) ARM: 11
  - iii) Endianness: 11
- b) Embedded Platforms: 11
  - i) Microcontroller: 11
  - ii) System on Chip: 11
  - iii) Single Board Computer: 11
  - iv) FPGA: 12
- c) Serial Communication: 12
  - i) SPI: 12
  - ii) I2C: 12
  - iii) UART: 12
  - iv) RS232: 13
  - v) RS485: 13
- d) GPIO: 13
  - i) Pin Modes: 14
  - ii) Direction: 14
- e) Analog Input: 14
  - i) ADC: 14
- f) Analog Output: 15
  - i) PWM: 15
  - ii) DAC: 15
- g) Timers: 15
- h) Constraints: 15
  - i) Power: 15
  - ii) Memory: 16

3) Embedded Operating Systems: 17
- a) Bare Metal: 17
- b) GNU/Linux: 17
- c) Real Time Operating System: 17

4) Embedded Software: 18
a) C Keywords: 18
i) volatile: 18
ii) static: 18
iii) const: 18
iv) pragma: 18
v) attribute: 18
vi) restrict: 18
vii) inline: 18

b) Watchdog Timer: 19
c) Memory Alignment: 19
d) Pointers: 19
e) Bit Manipulation: 19
i) Bitwise: 19
ii) Arithmetic: 19

f) Interrupts: 19
g) Optimizations: 20
h) Coding Standards: 20
i) MISRA: 20
ii) ISO/IEC90003: 20

5) Embedded Software Layer: 21
a) Peripheral and Hardware Layer: 21
i) Registers: 21
ii) Bit Fields: 21
iii) Memory Map: 21
iv) Peripherals: 21

b) Hardware Abstraction Layer (HAL): 21
i) Register Definition Files: 21
ii) Memory Access Methods: 21
iii) Board Support Package: 21

c) Application Layer: 22
i) Data Structures and Algorithms: 22
ii) Domain Knowledge: 22

6) Debugging: 23
a) Static Analysis: 24
b) Dynamic Analysis: 24
c) Logic Analyzers: 24
d) Oscilloscope: 24
e) JTAG: 24
f) SWD: 24

7) : Testing and Quality Management: 25

1) Development Environment:

a) Cross Compilations:

[ ] Article: "Compiling, Linking, and Locating", Barr Group

b) Development Tools:

i) IDE:

+ PlatformIO
+ Visual Studio Code
+ VisualGDB

ii) Oscilloscopes and Logic Analyzers:

[ ] Video: "Setting Up the Analog Discovery 2 Portable Lab Instrument", YouTube

2) Embedded Hardware:

[ ] Article: "Getting to Know the Hardware", Barr Group
[ ] Video: "Computer Hardware", YouTube
[ ] Video: "MSP430 Hardware Overview", YouTube

a) Computer Architecture:

[ ] Video: "Architecture-Software Interface", Coursera
[ ] Video: "Word Size and Data Types", Coursera

i) RISC (Reduced Instruction Set Computer):

[ ] Website "RISC-V Educational Materials", RISC-V

ii) ARM (Advanced RISC Machine):

[ ] Video: "ARM University, ARM Architecture Fundamentals"
[ ] Video: "Get to Know the ARM Cortex M7", YouTube

iii) Endianness:

[ ] Article: "Big Endian, Little Endian, Endianness: Understanding Byte Arrangements in Digital Systems", All About Circuits
[ ] Video: "Endianness", Coursera

b) Embedded Platforms:

i) Microcontroller:

[ ] PDF: "Introduction to Microcontrollers", Gunther Gridling, Bettina Weiss

ii) SoC (System on Chip):

[ ] Video: "Arm Education Media Launches System-on-Chip Design Online Courses", YouTube

iii) Single Board Computer:

[ ] Article: "Single-board computer", Wikipedia

iv) FPGA (Field Programmable Gate Array):

[ ] Article: "How FPGAs work, and why you'll buy one", Embedded Related
[ ] Video: "A Look Inside: SoC FPGAs Introduction (Part 1 of 5)", YouTube

c) Serial Communication:

[ ] Video: "PROTOCOLS: UART - I2C - SPI - Serial communications #001", YouTube

i) SPI:

[ ] Video "SPI - SPI Overview & Implementation on theMSP430", YouTube
[ ] Video: "SPI - Sending a Byte as a SPI Master", YouTube
[ ] Video: "SPI - Sending a Packet as a SPI Master using UCTXIFG", YouTube
[ ] Video: "SPI - Sending a Packet as a SPI Master using STE/SS", YouTube
[ ] Video: "SPI - Receiving a Byte as a SPI Master", YouTube
[ ] Video: "SPI - Slave Behavior", YouTube

ii) I2C:

[ ] Video: "I2C - What is I-Squared C and why the Resistors?"
[ ] Video: "I2C - Basic Packet Structure", YouTube
[ ] Video: "I2C - Addressing Slave Registers", YouTube
[ ] Video: "I2C - Master Configuration on the MSP430FR2355"
[ ] Video: "I2C - Adafruit PFC8523 Real-Time-Clock I2C Slave"
[ ] Video: "I2C - RTC-LaunchPad Connection & Making a Simple Probe", YouTube
[ ] Video: "I2C - Writing One Byte to an I2C Slave", YouTube
[ ] Video: "I2C - Writing a Register Addr + 3 Bytes to I2C Slave"
[ ] Video: "I2C - Reading One Byte from an I2C Slave", YouTube
[ ] Video: "I2C - Reading From a Specific Register Address"
[ ] Video: "I2C - Slave Operation", Youtube

iii) UART:

[ ] Video: "The UART - Serial Com Overview", YouTube
[ ] Video: "UART - The UART Standard", YouTube
[ ] Video: "UART - Configuring the UART Tx", YouTube
[ ] Video: "UART - Configuring the Baud Rate", YouTube
[ ] Video: "UART - Transmitting a Byte at 115200 Baud"
[ ] Video: "UART - Transmitting a Byte at 9600 Baud", YouTube
[ ] Video: "UART - Transmitting a Character to the Terminal"
[ ] Video: "UART - Transmitting a String to the Terminal"
[ ] Video: "UART - Transmitting String to the Terminal w/ IRQs"
[ ] Video: "UART - Configuring the UART Rx", YouTube
[ ] Video: "UART - Receiving Characters from the Terminal"

iv) RS232:

[ ] Video: "What is RS232 and What is it Used for?", YouTube
[ ] Video: "RS-232, RS-422, RS-485: What Are the Differences?", YouTube

v) RS485:

[ ] Video: "RS-485 Circuit Implementation", YouTube
[ ] Video: "What is RS485 and How it's used in Industrial Control Systems?", YouTube

d) GPIO:

[ ] Video: "How GPIO works | General Purpose Input Output | GPIO Behind The Scene", YouTube
[ ] Article: "Introduction to Microcontrollers - Hello World", Embedded Related
[ ] Article: "Introduction to Microcontrollers - More On GPIO", Embedded Related
[ ] Video: "Using GPIO with the MSP430 Microcontroller", YouTube

i) Pin Modes:

+ **Pull Up/Down:**

  [ ] Article: "Using Pull-Up and Pull-Down Resistors", Stratify Labs

+ **Push-Pull:**

  [ ] Video: "GPIO Output Configuration | Open Drain configuration | Push Pull configuration", YouTube

+ **Open Drain:**

  [ ] Video: "GPIO Output Mode: Working of Open Drain Configuration", YouTube

ii) Direction:

+ **Input:**

  [ ] Video: "MSP430 - Digital Inputs & Polling", YouTube

+ **Output:**

  [ ] Video: "MSP430 - Digital Outputs", YouTube

e) Analog Input:

i) ADC

[ ] Video: "Overview of ADCs", YouTube
[ ] Video: "The MSP430 ADC & Configuration", YouTube
[ ] Video: "Reading Voltage w/ Conversion-Complete Polling"
[ ] Video: "Reading Voltage w/ Conversion-Complete IRQ"
[ ] Video: "Reading Voltage w Conversion-Complete IRQ & LPM", YouTube

f) Analog Output:

i) PWM:

[ ] Video: "MSP430 - Creating PWM Signals using Timer Compares", YouTube

ii) DAC:

[ ] Article: "About Digital to Analog Converter (DAC) and Its Applications", ElProCus

g) Timers:

[ ] Article: "Introduction to Microcontrollers - More Timers and Displays", Embedded Related
[ ] Video: "Timer Overflow using ACLK", YouTube
[ ] Video: "Timer Overflow using ACLK + 12-Bit Counter Length"
[ ] Video: "Timer Overflow using SMCLK", YouTube
[ ] Video: "Timer Overflow using SMCLK + Divide-by-4 PreScalar"
[ ] Video: "Timer Compares", YouTube
[ ] Video: "Timer Captures", YouTube

h) Constraints:

[ ] Video: "Optimizing C for Microcontrollers - Best Practices - Khem Raj, Comcast RDK", YouTube

i) Power:

[ ] Video: "How To Lower AVR Microcontroller Power using Power Reduction Registers", YouTube
[ ] Video: "Atmel: picoPower Labs - Basic Power-Saving Techniques", YouTube
[ ] Video: "Microcontroller Design Considerations for Ultra Low Power Applications", YouTube
[ ] Video: "Ultra Low Power Microcontroller Design", YouTube

ii) Memory:

[ ] Article: "Memory", Barr Group
[ ] Video: "Interacting with Memory", Coursera
[ ] Video: "Different Types of Memory in Microcontroller : Flash Memory, SRAM and EEPROM", YouTube

+ **Program (ROM)**

  [ ] Video: "All about Microcontroller Memory", YouTube

+ **Dynamic (RAM)**

  [ ] Video: "Basics of Dynamic Memory Allocation", YouTube
  [ ] Article: "How to make a heap profiler", Embedded Related

3) Embedded Operating Systems:

[ ] Article: "Operating Systems", Barr Group

a) Bare Metal:

[ ] eBook: "Bare-metal C programming on ARM", GitHub
[ ] eBook: "Practical Guide to Bare Metal C++", GitBook

b) GNU/Linux:

[ ] ebook: "GNU Linux Rapid Embedded Programming", GitHub

c) Real Time Operating System:

[ ] Blog: "RTOS", Barr Group

4) Embedded Software:

[ ] Video: "Computer Software", YouTube
[ ] Video: "MSP430 Software Overview", YouTube

Important Programming Concepts (Even on Embedded Systems)

[ ] Article: "Part I: Idempotence", Embedded Related
[ ] Article: "Part II: Immutability", Embedded Related
[ ] Article: "Part III: Volatility", Embedded Related
[ ] Article: "Part IV: Singletons", Embedded Related
[ ] Article: "Part V: State Machines", Embedded Related
[ ] Article: "Part VI : Abstraction", Embedded Related

a) C Keywords:

[ ] Article: "Keywords to Frequent", Barr Group
[ ] Article: "Efficient C Code for 8-bit Microcontrollers", Barr Group

i) volatile:

[ ] Video: "How to use volatile keyword in C?", YouTube

ii) static:

[ ] Article: "Scope regions in C and C++", Embedded

iii) const:

[ ] Video: "C - const keyword", YouTube"

iv) pragma:

[ ] Article: "Pragmas", GCC GNU

v) attribute:

[ ] Video: "Compiler Attributes", Coursera
[ ] Article: "Using GNU C attribute", unixwiz

vi) restrict:

[ ] Video: "restric keyword in C", YouTube

vii) **inline:**

[ ] Article: "Inline Functions In C", greenend

b) Watchdog Timer:

[ ] Article: "Introduction to Watchdog Timers", Embedded

c) Memory Alignment:

[ ] Video: "Memory Alignment", Coursera

d) Pointers:

[ ] Video: "Pointers", Coursera

e) Bit Manipulation:

i) **Bitwise:**

[ ] Video: "Switching to C to Program the MSP430 - Bitwise Logic Operations", YouTube

ii) **Arithmetic:**

[ ] Video: "Switching to C to Program the MSP430 - Arithmetic"

f) Interrupts:

[ ] Article: "Introduction to Microcontrollers - Interrupts", Embedded Related
[ ] Video: "Overview and Basic Concepts", YouTube
[ ] Video: "Interrupts Overview & Port Interrupt Example"
[ ] Video: "Overview of the Interrupt Vector Table", YouTube
[ ] Video: "The use of the STACK during an IRQ + Nested IRQs"
[ ] Video: "The Responsibility of the Developer when using IRQs"
[ ] Video: "The IRQs on the MSP430FR2355 MCU", YouTube
[ ] Video: "Port Interrupts on the MSP430FR2355 MCU - Overview"
[ ] Video: "Reading from a Switch Using a Port IRQ Program Example"
[ ] Video: "Changing Edge Sensitivity (PxIES) & Forgetting to Clear a Flag", YouTube

g) Optimizations:

[ ] Article: "Optimizing Your Code", Barr Group

h) Coding Standards:

[ ] Article: "Embedded C Coding Standards", Barr Group
[ ] Video: "How Can I Enforce the SEI CERT C Coding Standard Using Static Analysis?", YouTube
[ ] Blog: "Coding Standards", Barr Group

i) MISRA:

[ ] Video: "An Introduction to MISRA C - Excerpt from An Introduction to MISRA C:2012 Webinar", YouTube

ii) ISO/IEC90003:

[ ] Article: "ISO/IEC/IEEE 90003:2018 Software engineering — Guidelines for the application of ISO 9001:2015 to computer software", ISO

5) Embedded Software Layer:

a) Peripheral and Hardware Layer:

[ ] Article: "Peripherals", Barr Group

i) Registers:

[ ] Video: "Level Up Your Arduino Code: Registers", YouTube

ii) Bit Fields:

[ ] Video: "Bit Fields in C. What are they, and how do I use them?", YouTube

iii) Memory Map:

[ ] Video: "Memory Map and Registers", Coursera

iv) Peripherals:

[ ] Video: "Exploring Configurable Logic Peripherals on PIC® and AVR® Microcontrollers", YouTube

b) Hardware Abstraction Layer (HAL):

[ ] Video: "The Nios® II Processor: Hardware Abstraction Layer"
[ ] Textbook: "Reusable Firmware Development: A Practical Approach to APIs, HALs and Drivers", PDFBOOK

i) Register Definition Files:

[ ] Video: "Register Definition Files", Coursera

ii) Memory Access Methods:

[ ] Video: "Direct Memory Access - DMA", YouTube

iii) Board Support Package:

[ ] Video: "0x1b7 What is a BSP | Board Support Package | Big Picture | Embedded Systems Software Development", YouTube

c) Application Layer:

i) Data Structures and Algorithms:

+ Queue:

- **Circular Buffer:**

  [ ] Video: Video: "Circular Buffer | Circular Buffer Implementation in C", YouTube

- **Semaphore:**

  [ ] Video: "Semaphores", YouTube
  [ ] Video: "What is difference between Semaphore and Mutex", YouTube

- **Event Handling:**

  [ ] Video: "Embedded Programming Lesson 33: Event-Driven Programming part-1", YouTube
  [ ] Video: Video: "Embedded Programming Lesson 34: Event-Driven Programming part-2", YouTube

- **Producer-Consumer:**

  [ ] Video: "Producer Consumer Pattern", YouTube
  [ ] Video: "Quick explanation: the Bounded-Buffer problem", YouTube

ii) Domain Knowledge:

+ Digital Signal Processing:

[ ] Video: "Learn Embedded Systems Design on ARM based Microcontrollers 1 of 2", YouTube
[ ] Video: "Learn DSP on ARM based Microcontrollers 2 of 2", YouTube

- **FFT:**

  [ ] Video: "Digital Signal Processing (DSP) Tutorial - DSP with the Fast Fourier Transform Algorithm"

- **Filters:**
  - **IIR:**

    [ ] Video: "IIR Filters - Audio DSP On STM32 with I2S (24 Bit / 96 kHz)", YouTube

  - **FIR:**

    [ ] Video: "FIR Filters - Audio DSP On STM32 (24 Bit / 48 kHz)", YouTube

+ **Control Systems:**

  - **PID:**

    [ ] Playlist: "Understanding PID Control", YouTube
    [ ] Playlist: "Drone Simulation and Control",

  - **Solar:**

    [ ] Video: "How to implement maximum power point tracking for solar charging", YouTube

+ **Cryptography and Encryption:**

  [ ] Video: "Embedded Cryptography", YouTube
  [ ] Playlist: "L36 Embedded Systems Security", YouTube

6) Debugging:

[ ] Article: "Downloading and Debugging", Barr Group
[ ] Blog: "Debugging", Barr Group

a) Static Analysis:

[ ] Video: "Static Code Analysis: Scan All Your Code For Bugs"
[ ] Video: "Using PC-Lint for MISRA and static code analysis", YouTube

b) Dynamic Analysis:

[ ] Video: "Static Analysis & Dynamic Testing of Software", YouTube

c) Logic Analyzers:

[ ] Video: "EEVblog #44 Part 1 - Logic Analyzer Tutorial", YouTube
[ ] Video: "EEVblog #44 - Part 2 - Logic Analyzer Tutorial", YouTube

d) Oscilloscope:

[ ] Video: "How to Debug Embedded Designs with an Oscilloscope"
[ ] Video: "How to test Automotive Serial Buses with Oscilloscopes"

e) JTAG:

[ ] Video: "EEVblog #449 - What is JTAG and Boundary", YouTube

f) SWD:

[ ] Video: "Learning Hardware SWD", YouTube

7) Testing and Quality Management:

Phil Koopman Lectures:

[ ] Playlist: "Embedded Software Testing", YouTube
[ ] Playlist: "Embedded Security, Safety, and Software Quality", YouTube

Conclusion

This road map provides you with a structured approach to learning and exploring the vast field of embedded systems engineering. By following the recommended resources and topics outlined here, you will gain a solid understanding of the fundamental concepts, tools, and techniques involved in this exciting discipline.

To further support your learning journey, I invite you to visit my open library on Google Drive. In this library, you will find an extensive collection of books, tutorials, and additional resources that can enhance your understanding and provide further opportunities for growth in the realm of embedded systems engineering. Feel free to access the library by clicking on the following link:
https://tinyurl.com/yanis-learning-folder

Remember, continuous learning and hands-on practice are key to mastering embedded systems engineering. As you progress through the road map and explore the resources provided, be sure to apply your knowledge through practical projects and real-world applications.

Wishing you an exciting and rewarding experience in your embedded systems engineering endeavors!

If you have any feedback or a suggestion, please leave a comment in this document!
Go to Insert Menu > Comment
or use the shortcut "Ctrl+Alt+M" or "Cmd+Options+M (on Mac)"

Visit this LINK to see the online version and leave your comment

If you want to be notified with updates, please leave your email in the comment section.

==========

Follow US

www.linktr.ee/electro.sc

==========

Please always refer to this LINK in order to get the latest update to this doc!
Check out my resources HERE!

If you want to be notified with updates, please leave your email in the comment section.